Contents

Chapter 2 — The Axioms of Probability

2-1 Set Theory

Sets and Subsets

- **Set**: A collection of objects called elements. A set A consisting of elements $\zeta_1, \dots, \zeta_n$ is denoted $A = \{\zeta_1, \dots, \zeta_n\}$.
- **Subset**: $B \subset A$ means every element of B is also an element of A. All sets under consideration are subsets of a universal space S.
- **Empty Set**: Denoted by $\{\emptyset\}$, it contains no elements.
- **Partitions**: A partition U of a set S is a collection of mutually exclusive subsets A_i whose union equals S.

Set Operations

- **Union (Sum)**: $A \cup B$ (or $A + B$) is the set of all elements in A or B or both.
- **Intersection (Product)**: $A \cap B$ (or AB) consists of elements common to both A and B.
- **Mutually Exclusive**: Sets A and B are mutually exclusive if they have no common elements, i.e., $AB = \{\emptyset\}$.
- **Complement**: $\overline{A}$ consists of all elements in S not in A. Note that $A\overline{A} = \{\emptyset\}$ and $A \cup \overline{A} = S$.

Important Laws

- **De Morgan's Law**: $\overline{A \cup B} = \overline{A} \cap \overline{B}$.
- **Duality Principle**: If in a set identity we replace all unions by intersections, all intersections by unions, and the sets S and $\{\emptyset\}$ by $\{\emptyset\}$ and S, the identity is preserved.

2-2 Probability Space

In probability theory, S is the **certain event**, its elements are **experimental outcomes**, its subsets are **events**, and $\{\emptyset\}$ is the **impossible event**.

The Axioms of Probability

We assign to each event A a number $P(A)$, called the probability of the event, satisfying three conditions: 1. **Non-negativity**: $P(A) \geq 0$. 2. **Certainty**: $P(S) = 1$. 3. **Additivity**: If $AB = \{\emptyset\}$, then $P(A \cup B) = P(A) + P(B)$.

Corollaries and Properties

- **Impossible Event**: $P(\{\emptyset\}) = 0$.
- **Complement Probability**: For any A, $P(\overline{A}) = 1 - P(A) \leq 1$.
- **General Union Rule**: For any A and B,

$$P(A \cup B) = P(A) + P(B) - P(AB) \leq P(A) + P(B)$$

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- **Subset Probability**: If $B \subset A$, then $P(A) = P(B) + P(A\overline{B}) \geq P(B)$.

Fields and Borel Fields

- **Field ($\mathcal{F}$)**: A nonempty class of sets where if $A \in \mathcal{F}$, then $\overline{A} \in \mathcal{F}$, and if $A \in \mathcal{F}$ and $B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$. A field always contains S and $\{\emptyset\}$.
- **Borel Field**: If the union and intersection of an infinite sequence of sets $A_1, \ldots, A_n, \ldots$ in $\mathcal{F}$ also belongs to $\mathcal{F}$, then $\mathcal{F}$ is a Borel field.

2-3 Conditional Probability

Definition and Properties

The conditional probability of an event A assuming another event M is defined as:

$$P(A|M) = \frac{P(AM)}{P(M)}$$

where $P(M) > 0$.

- **Subset Conditions**:
  - If $M \subset A$, then $P(A|M) = 1$.
  - If $A \subset M$, then $P(A|M) = \frac{P(A)}{P(M)} \geq P(A)$.
- **Axiom Compliance**: Conditional probabilities are valid probabilities; they satisfy the axioms. Specifically, $P(A \cup B|M) = P(A|M) + P(B|M)$ if A and B are mutually exclusive.

Example 2-10 (Die Experiment): We want the conditional probability of rolling a 2, given that an even number was rolled. Let $A = \{f_2\}$ and $M = \{\text{even}\} = \{f_2, f_4, f_6\}$. $P(A) = 1/6$, $P(M) = 3/6$, and $AM = A$. Thus:

$$P(f_2|\text{even}) = \frac{P(\{f_2\})}{P(\text{even})} = \frac{1/6}{3/6} = \frac{1}{3}$$

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Example 2-13 (Drawing Balls without Replacement): A box contains 3 white and 2 red balls. Two balls are drawn in succession. To find the probability that the first is white (W_1) and the second is red (R_2): $P(W_1) = 3/5$. If a white ball is removed, 2 white and 2 red balls remain, so $P(R_2|W_1) = 2/4$. Using the conditional probability definition:

$$P(W_1 R_2) = P(R_2|W_1)P(W_1) = \left(\frac{2}{4}\right)\left(\frac{3}{5}\right) = \frac{6}{20}$$

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Independent Events

Two events A and B are said to be independent if their joint probability equals the product of their individual probabilities:

$$P(AB) = P(A)P(B)$$

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Would you like me to extract any specific proofs or additional homework problems from this chapter?